

Monte Carlo Methods

Introduction

Generating Samples

Markov Chain Monte Carlo

Improving Efficiency

Selected Topics

Git website and repository

Canvas

Fred Hickernell, Fall 2025

Updated 2025 September 3



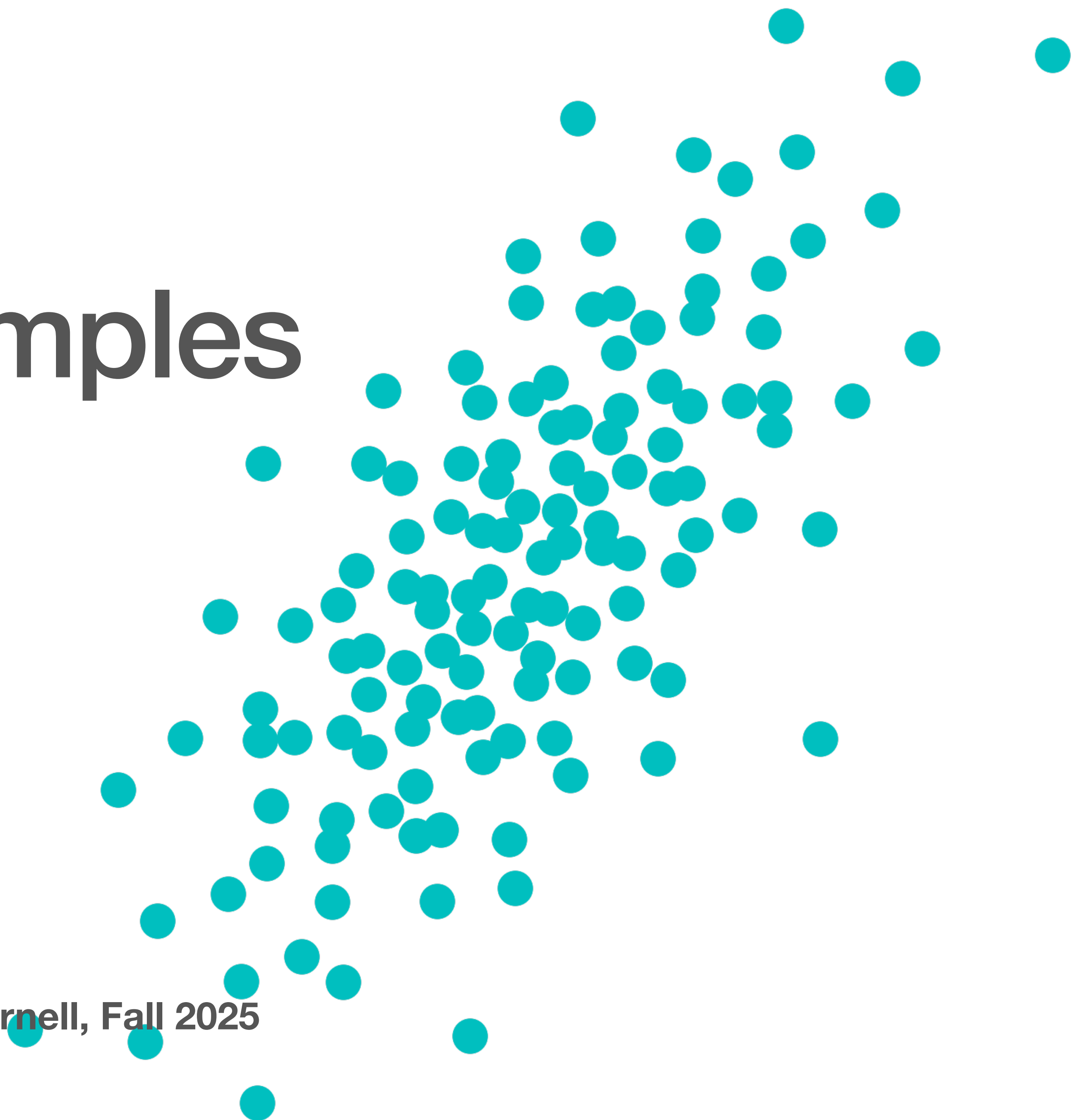
Generating Samples

Owen, Chapters 3 – 6

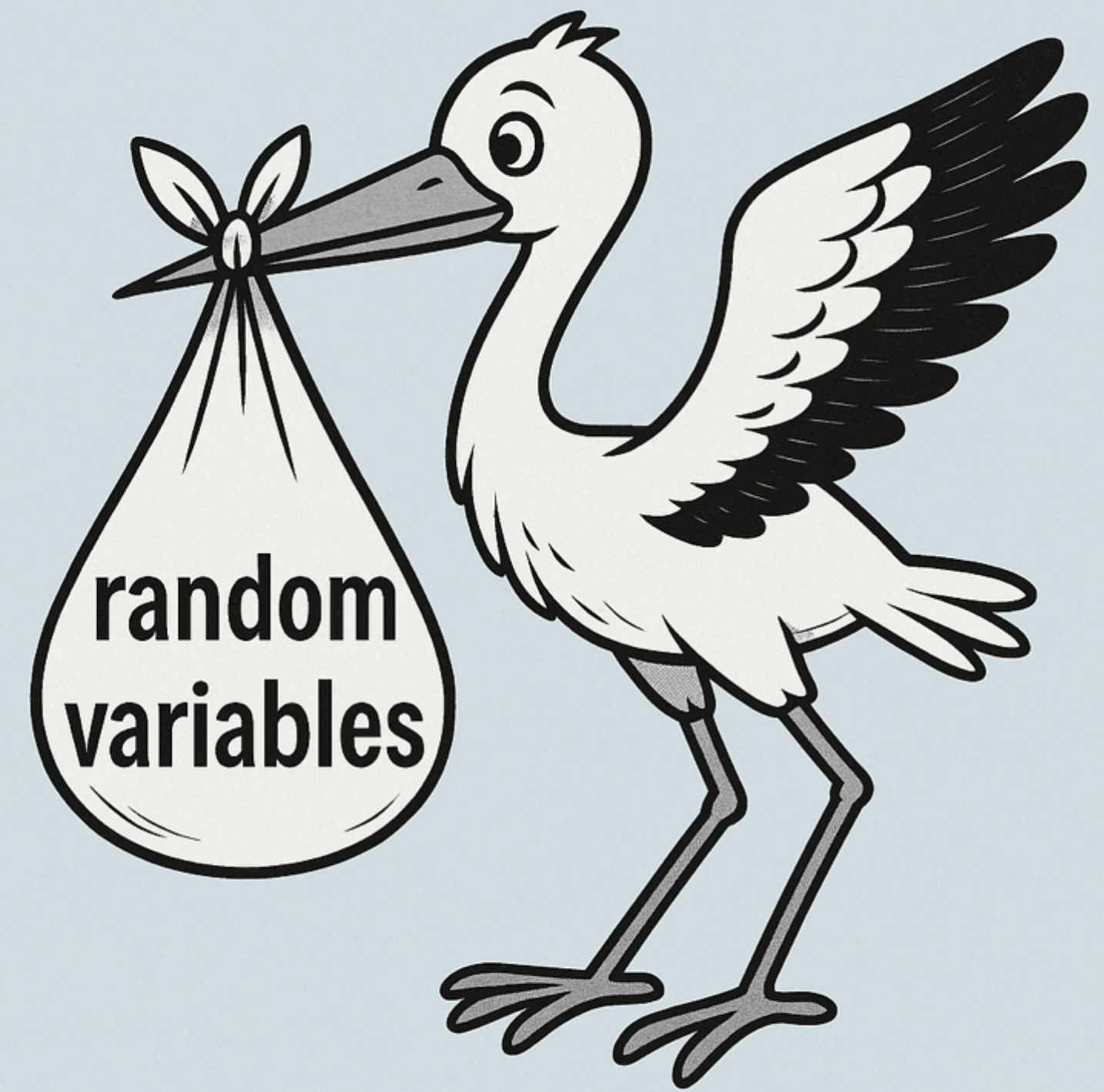
Project Selection due Oct 1

Assignment ?

MATH 565 Monte Carlo Methods, Fred Hickernell, Fall 2025



Where do random variables come from?





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Where do random variables come from?

- Algorithms generate (typically **uniform pseudo-random** numbers
- Pseudo-random number generators are tested to ensure that their output **looks IID random** in every way possible
- Care must be taken when working on **multiple processors**
- For a drastic failure see [RANDU](#)
- **Physical** random number generators may be random, but cannot be guaranteed to be IID random of the desired distribution





Quantile function gives non-uniform random numbers

- Most random number generators output $U_1, U_2, \dots$ that mimic IID $\mathcal{U}[0,1]$.
- To get $X_1, X_2, \dots$ IID with cumulative distribution function F , we use the quantile function, Q , where $Q(u) := \inf\{x \in \mathcal{X} : F(x) \geq u\}$. Note that

$$Q(u) \leq x \iff u \leq F(x)$$

- Letting $X := Q(U)$, it follows that

$$\mathbb{P}(X \leq x) = \mathbb{P}(Q(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x)$$

and so $X \sim F$, and

$$X_1 := Q(U_1), X_2 := Q(U_2), \dots \stackrel{\text{IID}}{\sim} F$$



Binomial random numbers



Binomial random numbers

Suppose $X \sim \text{Binomial}(3, 0.6)$



Binomial random numbers

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x	0	1	2	3
PMF $p(x)$	0.4^3 0.064	$3(0.6)(0.4^2)$ 0.288	$3(0.6^2)(0.4)$ 0.432	0.6^3 0.216
$x \in$	$[0, 1)$	$[1, 2)$	$[2, 3)$	$[3, \infty)$
CDF $F(x)$	0.064	0.352	0.784	1
$u \in$	$(0, 0.064]$	$(0.064, 0.352]$	$(0.352, 0.784]$	$(0.784, 1]$
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Note that F is **right**-continuous and Q is **left**-continuous



Explorations in generating random vectors

Generating Samples



Zero-inflated exponential

Suppose that X is a non-negative random variable with probability p_0 of being zero and otherwise an exponential distribution

$$F(x) = \begin{cases} 0 & -\infty < x < 0 \\ p_0 + (1 - p_0)[1 - \exp(-\lambda x)] & 0 \leq x < \infty \end{cases}$$

To generate $X_1, X_2, \dots \stackrel{\text{IID}}{\sim} F$ from $U_1, U_2, \dots \stackrel{\text{IID}}{\sim} \mathcal{U}[0,1]$, we need the **quantile function**



Quantile function for zero-inflated exponential

$$F(x) = \begin{cases} 0 & -\infty < x < 0 \\ p_0 + (1 - p_0)[1 - \exp(-\lambda x)] & 0 \leq x < \infty \end{cases}$$

$$u = F(x) \implies \begin{cases} x = 0, & 0 < u \leq p_0 \\ u = p_0 + (1 - p_0)[1 - \exp(-\lambda x)], & p_0 \leq u < 1 \end{cases}$$

$$\iff \begin{cases} x = 0, & 0 < u \leq p_0 \\ \frac{u - p_0}{1 - p_0} = 1 - \exp(-\lambda x), & p_0 \leq u < 1 \end{cases}$$

$$\iff x = Q(u) = \begin{cases} 0, & 0 < u \leq p_0 \\ -\frac{1}{\lambda} \log \left(1 - \frac{u - p_0}{1 - p_0} \right), & p_0 \leq u < 1 \end{cases}$$



Random vectors with **independent** marginals

Suppose that $X = (X_1, \dots, X_d)$ is a random **vector** with independent marginals. Then

$$X = (Q_1(U_1), \dots, Q_d(U_d)), \quad U \sim \mathcal{U}[0,1]^d$$

has the desired distribution, provided that $Q_1, \dots, Q_d$ are the corresponding **quantile functions**.

We express n samples of X as an $n \times d$ matrix or array,

$$X = \begin{pmatrix} X_{11} & X_{12} & \cdots & X_{1d} \\ \vdots & \vdots & & \vdots \\ X_{n1} & X_{n2} & \cdots & X_{nd} \end{pmatrix}, \text{ where } X_{ij} \text{ is the } j^{\text{th}} \text{ **component** of the } i^{\text{th}} \text{ **sample**}$$



What is your expertise in computing and probability?

- Go to menti.com
- Use code 4845 0474



Multivariate normal distribution

If $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}_d, \mathbf{I}_d)$, i.e., a d -dimensional standard normal random variable, then $\mathbf{X} = \mathbf{A}\mathbf{Z} + \mathbf{b}$ (thinking of $\mathbf{Z}$ as a column vector) has

$$\boldsymbol{\mu} = \mathbb{E}(\mathbf{X}) = \mathbf{b}$$

$$\begin{aligned}\Sigma &= \text{cov}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top] \\ &= \mathbb{E}[\mathbf{A}\mathbf{Z}\mathbf{Z}^\top\mathbf{A}^\top] = \mathbf{A}\mathbf{A}^\top\end{aligned}$$

So $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, where Σ is **symmetric** and **positive-definite**. So, give a desired $\boldsymbol{\mu}$ and Σ , one needs only to find $\mathbf{A}$ with $\Sigma = \mathbf{A}\mathbf{A}^\top$, and then

$$\mathbf{X} = \mathbf{A}\mathbf{Z} + \boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

Note that there are **multiple** ways to decompose $\Sigma = \mathbf{A}\mathbf{A}^\top$.



Cholesky decomposition of Σ

We want to find a lower triangular matrix A , such that $\Sigma = AA^\top$.

$$\begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix} = \Sigma = AA^\top = \begin{pmatrix} a_{11} & 0 \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} a_{11} & a_{21} \\ 0 & a_{22} \end{pmatrix}$$

$$= \begin{pmatrix} a_{11}^2 & a_{11}a_{21} \\ a_{11}a_{21} & a_{21}^2 + a_{22}^2 \end{pmatrix}$$

$$\implies a_{11} = \sqrt{3}$$

$$a_{21} = -1/a_{11} = -1/\sqrt{3}$$

$$a_{22} = \sqrt{1 - a_{21}^2} = \sqrt{2/3}$$

$$A = \begin{pmatrix} \sqrt{3} & 0 \\ -1/\sqrt{3} & \sqrt{2/3} \end{pmatrix}$$



Gaussian processes

- Random **functions** such that evaluating them at a finite number of points gives a random vector with a multivariate normal distribution
- The distribution is denoted $\mathcal{GP}(\mu, K)$, where μ is the **mean** function and K is the **covariance** kernel
- $g \sim \mathcal{GP}(\mu, K) \implies$
 - $\mathbb{E}[g(t)] = \mu(t)$
 - $\mathbb{E}[\{g(t) - \mu(t)\}\{g(x) - \mu(x)\}] = K(t, x)$



Brownian Motion

Special case of a Gaussian process

$$\mu(t) = 0 \quad \text{and} \quad K(t, x) = \min(t, x), \quad 0 \leq t$$

For $0 \leq t_1 < t_2 < \dots \leq t_d$,

$$X = (B(t_1), \dots, B(t_d))^{\top} \sim \mathcal{N}(\mathbf{0}, \Sigma)$$

where

$$\Sigma = \begin{pmatrix} t_1 & t_1 & \dots & t_1 \\ t_1 & t_2 & \dots & t_2 \\ \vdots & \vdots & \ddots & \vdots \\ t_1 & t_2 & \dots & t_d \end{pmatrix} = \begin{pmatrix} \sqrt{t_1} & 0 & \dots & 0 \\ \sqrt{t_1} & \sqrt{t_2 - t_1} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \sqrt{t_1} & \sqrt{t_2 - t_1} & \dots & \sqrt{t_d - t_{d-1}} \end{pmatrix} \begin{pmatrix} \sqrt{t_1} & \sqrt{t_1} & \dots & \sqrt{t_1} \\ 0 & \sqrt{t_2 - t_1} & \dots & \sqrt{t_2 - t_1} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sqrt{t_d - t_{d-1}} \end{pmatrix}$$



Brownian Motion

For this Cholesky decomposition of the covariance matrix

$$\Sigma = \begin{pmatrix} t_1 & t_1 & \cdots & t_1 \\ t_1 & t_2 & \cdots & t_2 \\ \vdots & \vdots & \ddots & \vdots \\ t_1 & t_2 & \cdots & t_d \end{pmatrix} = \begin{pmatrix} \sqrt{t_1} & 0 & \cdots & 0 \\ \sqrt{t_1} & \sqrt{t_2 - t_1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \sqrt{t_1} & \sqrt{t_2 - t_1} & \cdots & \sqrt{t_d - t_{d-1}} \end{pmatrix} \begin{pmatrix} \sqrt{t_1} & \sqrt{t_1} & \cdots & \sqrt{t_1} \\ 0 & \sqrt{t_2 - t_1} & \cdots & \sqrt{t_2 - t_1} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sqrt{t_d - t_{d-1}} \end{pmatrix}$$

One may construct, IID random samples $X_i = (B(t_1), \dots, B(t_d))^T \stackrel{\text{IID}}{\sim} \mathcal{N}(\mathbf{0}, \Sigma)$ by

$$X_{i1} = \sqrt{t_1} Z_{i1}, \quad X_{i2} = X_{i1} + \sqrt{t_2 - t_1} Z_{i2}, \quad \dots, \quad X_{id} = X_{i,d-1} + \sqrt{t_d - t_{d-1}} Z_{id}$$

where $Z_i \stackrel{\text{IID}}{\sim} \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$



Stock/Asset Prices

Brownian motions are used in financial modeling. If

- S_0 is the initial price of a stock/asset
- r is the interest rate
- σ is the volatility

Then

$$S(t) = S_0 \exp\left((r - \sigma^2/2)t + \sigma B(t)\right)$$

is a **geometric Brownian motion** model of the **random** stock price



Option Payoffs and Prices

Give some model of asset prices, we may model the payoff of an option.

Let K be the **strike price** and T be the time to maturity. Examples of **discounted payoffs** are

European call	$\max(S(T) - K, 0) \exp(-rT)$
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European put	$\max(K - S(T), 0) \exp(-rT)$
--------------	-------------------------------

Asian Arithmetic mean call	$\max\left(\frac{1}{T} \int_0^T S(t) dt - K, 0\right) \exp(-rT)$
----------------------------	--

The fair price of an option is

$$\mu = \mathbb{E}[\text{payoff}]$$



Low discrepancy sampling

If we are willing to accept samples that are not IID, then we may get substantial gains in the convergence rate. We say that

$$U_1, \dots, U_n \stackrel{\text{LD}}{\sim} \mathcal{U}[0,1]^d$$

if the **empirical distribution** of $\{U_1, \dots, U_n\}$ is close to the uniform distribution. The measure of how far away these distributions are is the **discrepancy**, and it will be defined precisely later. Some examples are

- Digital sequences, especially Sobol' sequences
- Lattice sequences
- Halton sequences
- Kronecker sequences

These come in **deterministic** and **random** forms.



Low discrepancy sampling

- LD sequences are available through [scipy](#) and [qmcpy](#), which has been developed in our research group
- Rows correspond to samples and columns to coordinates
- See [Generating Samples](#) for empirical comparisons
- The theory of LD sampling will be touched on later in the semester



Bias–variance decomposition

- Let μ be any population quantity, e.g., the mean
- Let $\hat{\mu}$ be any estimator for μ , e.g., the sample mean
- Then the mean squared error of $\hat{\mu}$ is

$$\begin{aligned}\text{mse}(\hat{\mu}) &= \mathbb{E}[(\mu - \hat{\mu})^2] = \mathbb{E}[\{\mu - \mathbb{E}(\hat{\mu})\} + \{\mathbb{E}(\hat{\mu}) - \hat{\mu}\}]^2 \\ &= \mathbb{E}[\{\mu - \mathbb{E}(\hat{\mu})\}^2] + \mathbb{E}[\{\mathbb{E}(\hat{\mu}) - \hat{\mu}\}^2] + 2\mathbb{E}[\{\mu - \mathbb{E}(\hat{\mu})\}\{\mathbb{E}(\hat{\mu}) - \hat{\mu}\}] \\ &= [\text{bias}(\hat{\mu})]^2 + \text{var}(\hat{\mu}), \quad \text{where } \text{bias}(\hat{\mu}) := \mu - \mathbb{E}(\hat{\mu})\end{aligned}$$

IID sample means have **zero bias** and positive variance

Deterministic LD sample means have positive bias but **zero variance**



Acceptance–rejection sampling

Suppose that

- You **can** sample a random variable Z with known probability density q_Z
- You really **want to** sample X with known (unnormalized) probability density q , i.e., the true density is cq for some positive constant c
- And we **know** a M for which $q(x) \leq Mq_Z(x)$ for all x

The acceptance rejection sampling proceeds as follows

```
i ← 0
repeat
  repeat
    generate  $Z \sim q_Z$  and  $U \sim \mathcal{U}[0,1]$ 
  until  $U \leq q(Z) / [cq_Z(Z)]$ 
   $X[i] \leftarrow Z$ ,  $i \leftarrow i+1$ 
until  $i=n$ 
```




Why does acceptance–rejection sampling work?

$$W := \begin{cases} 1, & U \leq \varrho(Z)/[M\varrho_Z(Z)] \\ 0, & \text{otherwise} \end{cases} \quad \begin{array}{l} \text{accept} \\ \text{reject} \end{array}$$

Our sampling method generates X that has a PDF of

$\tilde{\varrho}(z) = \varrho_{Z|W}(z|1) = \text{PDF of } Z \text{ conditioned on accepting } Z$

$$= \frac{\varrho_{W|Z}(1|z)\varrho_Z(z)}{\varrho_W(1)}$$

by Bayes Theorem

$$= \frac{[M\varrho(z)/\varrho_Z(z)]\varrho_Z(z)}{\varrho_W(1)}$$

by our acceptance rule

$$= \frac{M\varrho(z)}{\varrho_W(1)} = c\varrho(z) \text{ since } c\varrho \text{ is a density, so probability of acceptance is } M/c$$

To get n samples of X , we need on average nc/M samples of Z



Explorations in acceptance-rejection

Acceptance Rejection Sampling



Comparing Methods for Non-Uniform Sampling

Method	Requirements	LD friendly
$X = Q(U)$	quantile function	Yes
$X = AZ + \mu \sim \mathcal{N}(\mu, \Sigma)$	$\Sigma = AA^T$	Yes
acceptance-rejection	$\varrho(x) \leq c\varrho_Z(z)$	Somewhat

For acceptance-rejection with LD see [\[Zhu & Dick 2024\]](#)
This is a good project topic